

STRONGER MULTIPLICITY ONE FOR SELBERG'S CLASS

*M. Ram Murty*¹

1. Introduction.

The Selberg class $\mathcal{S}$ consists of functions $F(s)$ of a complex variable s satisfying the following properties:

- (i) (Dirichlet series): for $\operatorname{Re}(s) > 1$,

$$F(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

where $a_1 = 1$. (We will write $a_n(F) = a_n$ for the coefficients of the Dirichlet series of F .)

- (ii) (Analytic continuation): for some integer $m \geq 0$, $(s-1)^m F(s)$ extends to an entire function of finite order.
- (iii) (Functional equation): there are numbers $Q > 0$, $\alpha_i > 0$, $r_i \in \mathbb{C}$ with $\operatorname{Re}(r_i) \geq 0$ so that

$$\Phi(s) = Q^s \prod_{i=1}^d \Gamma(\alpha_i s + r_i) F(s)$$

satisfies the functional equation

$$\Phi(s) = w \overline{\Phi}(1-s)$$

where w is a complex number with $|w| = 1$ and $\overline{\Phi}(s) = \overline{\Phi(\overline{s})}$.

- (iv) (Euler product) for $\operatorname{Re}(s) > 1$,

$$F(s) = \prod_p F_p(s)$$

where

$$F_p(s) = \exp \left(\sum_{k=1}^{\infty} \frac{b_{p^k}}{p^{ks}} \right)$$

and $b_{p^k} = O(p^{k\theta})$ for some $\theta < 1/2$ and p denotes a prime number here and throughout this paper. We shall write $b_p(F) = b_p$.

- (v) (Ramanujan hypothesis) for any fixed $\epsilon > 0$,

$$a_n = O(n^\epsilon).$$

¹ *Research partially supported by NSERC, FCAR and CICMA grants.*

A prototypical example of an element of $\mathcal{S}$ is of course the Riemann zeta function. But more exemplary is the Ramanujan zeta function

$$L_{\Delta}(s) = \sum_{n=1}^{\infty} \frac{\tau_n}{n^s}$$

where $\tau_n = \tau(n)/n^{1/2}$ and the Ramanujan τ -function is defined by the infinite product:

$$\sum_{n=1}^{\infty} \tau(n)q^n = q \prod_{n=1}^{\infty} (1 - q^n)^{24}$$

Ramanujan established the analytic continuation and functional equation for $L_{\Delta}(s)$ and conjectured that (iv) and (v) also hold. These conjectures, as is well-known, were proven by Mordell and Deligne respectively.

A function $F \in \mathcal{S}$ is called primitive if it cannot be written as a non-trivial product of two elements of $\mathcal{S}$. By considering the integral

$$\frac{1}{2\pi i} \int_{R_T} \frac{F'}{F}(s) ds$$

with R_T the contour described by $-\delta \leq \operatorname{Re} s \leq 1 + \delta$ (with δ sufficiently small) and $0 \leq \operatorname{Im} s \leq T$ and using Stirling's formula and the functional equation, one derives easily an asymptotic formula for the number of zeroes of $F(s)$ in R_T . Let us denote this by $N_F(T)$. Then,

$$N_F(T) \sim \left(2 \sum_{i=1}^d \alpha_i \right) \frac{T \log T}{2\pi}$$

as $T \rightarrow \infty$. We therefore see that even though the α_i are not well-defined in the functional equation (by virtue of Legendre's duplication formula), the sum of the α_i 's is well-defined. Accordingly, we define the dimension of F as

$$\dim F = 2 \sum_{i=1}^d \alpha_i$$

Conrey and Ghosh [CG] have shown that if $\dim F = 0$, then $F = 1$ (essentially from condition (iv) defining $\mathcal{S}$). Bochner [B] proved that if $\dim F > 0$, then $\dim F \geq 1$. We briefly indicate the method of his proof below. Consider the identity,

$$\sum_{n=1}^{\infty} a_n e^{-nx} = \frac{1}{2\pi i} \int_{2-i\infty}^{2+i\infty} F(s) x^{-s} \Gamma(s) ds$$

Because of the Phragmen-Lindelöf principle, and the functional equation, we find that $F(s)$ has polynomial growth in $|\operatorname{Im}(s)|$ in any vertical strip. Thus, moving the line of

integration to the left, and taking into account the possible pole at $s = 1$ of $F(s)$ as well as the poles of $\Gamma(s)$ at $s = 0, -1, -2, \dots$ we find

$$\sum_{n=1}^{\infty} a_n e^{-nx} = \frac{P(\log x)}{x} + \sum_{n=0}^{\infty} \frac{F(-n)(-1)^n}{n!} x^n$$

where P is a polynomial. The functional equation relates $F(-n)$ to $F(n+1)$ with a product of gamma functions. If $0 < \dim F < 1$, we find by Stirling's formula that the sum on the right hand side of the above identity converges for all x (including complex values of x). $P(\log x)$ is analytic in $\mathbb{C} \setminus \{x \leq 0 : x \in \mathbb{R}\}$. Hence the left hand side is analytic in

$$\mathbb{C} \setminus \{x \leq 0 : x \in \mathbb{R}\}$$

But since the left hand side is periodic with period $2\pi i$, we find that

$$f(z) = \sum_{n=1}^{\infty} a_n e^{-nz}$$

is entire. Thus, we find that for any x ,

$$a_n e^{-nx} = \int_0^{2\pi} f(x + iy) e^{-iny} dy \\ \ll n^{-2}$$

by integrating by parts. Choosing $x = 1/n$ we find $a_n = O(n^{-2})$. Hence the Dirichlet series

$$F(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}$$

converges absolutely for $\operatorname{Re} s > -1$. However, relating $F(-\frac{1}{2} + it)$ to $F(\frac{3}{2} - it)$ by the functional equation and using Stirling's formula, we find that $F(-\frac{1}{2} + it)$ is not bounded. This proves $\dim F \geq 1$.

We immediately deduce that any function $F \in \mathcal{S}$, $F \neq 1$ can be written as a product of primitive functions. One of the important consequences of Selberg's conjectures, to be described below, is that this factorization is unique. Such a unique factorization theorem would have important consequences to Artin's conjecture and the Langlands' reciprocity conjecture ([M1]).

Thus, $\zeta(s)$ and Dirichlet's L -functions $L(s, \chi)$ are primitive functions of dimension 1. In [M2], it was shown that $L_{\Delta}(s)$ and more generally, an L -function attached to a newform is primitive. If the Fourier coefficients of Maass forms satisfy the Ramanujan conjecture, then their associated L -functions lie in the Selberg class and one can again show that these are primitive of dimension 2.

It is conjectured that $\dim F \in \mathbb{Z}$. It is still an open problem to classify primitive functions of dimension 1. If F has dimension 1, and the α_i appearing in the functional

equation of F are rational numbers, then F is either $\zeta(s)$ or $L(s+it, \chi)$ for some Dirichlet character χ and $t \in \mathbb{R}$. This is essentially deduced from a theorem of Bochner [B]. (See [M2] and the discussion there.)

One suspects that $\mathcal{S}$ coincides with the family of automorphic L -functions, but this would be very difficult to establish.

2. Selberg's conjectures

Selberg [S] conjectures that for $x \rightarrow \infty$:

(a) for any primitive function F ,

$$\sum_{p \leq x} \frac{|a_p(F)|^2}{p} = \log \log x + O(1) :$$

(b) for two distinct primitive functions F and G ,

$$\sum_{p \leq x} \frac{a_p(F) \overline{a_p(G)}}{p} = O(1).$$

It is easy to deduce that (a) and (b) imply unique factorization into primitive functions. It is also easy to see that if F and G are elements of $\mathcal{S}$ (which are not necessarily primitive) and $a_p(F) = a_p(G)$ for all but finitely many primes p , then $F = G$. Indeed, let

$$\begin{aligned} F &= F_1^{e_1} \dots F_r^{e_r} \\ G &= F_1^{f_1} \dots F_r^{f_r} \end{aligned}$$

with $F_1, \dots, F_r$ primitive functions, e_i, f_i non-negative integers. We want to show $e_i = f_i$ for all i . Without loss of generality, suppose $e_1 \neq f_1$. Then

$$(1) \quad \sum_i e_i a_p(F_i) = \sum_i f_i a_p(F_i)$$

so that

$$\begin{aligned} & e_1 \sum_{p \leq x} \frac{|a_p(F_1)|^2}{p} + \sum_{i \geq 2} e_i \left(\sum_{p \leq x} \frac{a_p(F_i) \overline{a_p(F_1)}}{p} \right) \\ &= f_1 \sum_{p \leq x} \frac{|a_p(F_1)|^2}{p} + \sum_{i \geq 2} f_i \left(\sum_{p \leq x} \frac{a_p(F_i) \overline{a_p(F_1)}}{p} \right) \end{aligned}$$

upon multiplying both sides of equation (1) by $\overline{a_p(F_1)}/p$ and then summing over $p \leq x$. Now (a) and (b) imply that

$$e_1 \log \log x + O(1) = f_1 \log \log x + O(1)$$

as $x \rightarrow \infty$. Hence $e_1 = f_1$, as desired.

In analogy with the theory of automorphic L -functions, we therefore see that the Selberg class should have the “strong multiplicity one” property. That is, $a_p(F) = a_p(G)$ for all but finitely many primes p , implies $F = G$. In [MM], a weaker version of this theorem was proved: namely, if $a_p(F) = a_p(G)$ and $a_{p^2}(F) = a_{p^2}(G)$, for all but finitely many primes p , then $F = G$. In particular, if $F_p(s) = G_p(s)$ for all but finitely many primes p , then $F = G$.

The goal of this paper is to prove the strong multiplicity one property itself. Namely, we would like to show that if $a_p(F) = a_p(G)$ for all but finitely many primes p , then $F = G$. We are unable to do this unconditionally and must impose the following condition:

The Quadratic Twist Hypothesis: for any $F \in \mathcal{S}$, and any fundamental discriminant D , there is an element $F_D \in \mathcal{S}$ such that

$$F_D(s) = \sum_{n=1}^{\infty} \left(\frac{D}{n} \right) \frac{a_n}{n^s}$$

for $\operatorname{Re}(s) > 1$. Moreover, given $F, G \in \mathcal{S}$, there is a D such that

$$\operatorname{ord}_{s=1/2} F_D(s) = \operatorname{ord}_{s=1/2} G_D(s)$$

We make some remarks on the naturalness of the hypothesis. First, the existence of quadratic twists of elements of $\mathcal{S}$ is a special case of a more general conjecture of Selberg that $\mathcal{S}$ is closed under “twists by Dirichlet characters”. More precisely, if $F \in \mathcal{S}$, and χ is a primitive Dirichlet character, then there is an element $F_\chi \in \mathcal{S}$ such that

$$F_\chi(s) = \sum_{n=1}^{\infty} \frac{a_n \chi(n)}{n^s}$$

for $\operatorname{Re}(s) > 1$.

The second part of the hypothesis dealing with orders of zeroes of quadratic twists is also natural. Even in some simple special cases, it remains unsolved. For instance, suppose we have two newforms f, g of weight 2 and level N . Can we find a quadratic twist such that

$$\sum_{n=1}^{\infty} \frac{a_n(f)}{n} \left(\frac{D}{n} \right) = \sum_{n=1}^{\infty} \frac{a_n(g)}{n} \left(\frac{D}{n} \right) \neq 0$$

For this, one would need to study the asymptotic behaviour of sums like

$$\sum_{D \leq y} L(f, \chi_D, 1) L(g, \chi_D, 1)$$

where we are writing χ_D for $\left(\frac{D}{\cdot} \right)$ and

$$f(z) = \sum_{n=1}^{\infty} a_n(f) e^{2\pi i n z}$$

is the Fourier expansion of the cusp form.

(Note that we have not normalized $L(f, \chi_D, s)$ above so that the functional equation of $L(f, \chi_D, s)$ relates the value at s to $1 - s$. The functional equation relates the value at s to the value at $2 - s$ and so the central point of symmetry is $s = 1$).

One could perhaps employ the method of [MM2] to derive asymptotic formulae for sums of the above type. This would involve a delicate treatment to ensure that the error terms can be controlled. Even the case when F is a newform and G corresponds to a classical Dirichlet series associated to a quadratic character is a new problem worthy of careful investigation.

3. The explicit formula method.

The main reference to this section is Chapter 17 of Lang [L]. Let $F \in \mathcal{S}$. We write

$$\frac{-F'}{F}(s) = \sum_{p,k} \frac{b_{p^k}(F)}{p^{ks}} \log p.$$

Now suppose $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfies the following conditions:

- (i) there is an $\epsilon > 0$ such that $f(x) \exp((1 + \epsilon)x)$ is integrable and of bounded variation;
- (ii) the function $(f(x) - f(\phi))/x$ is of bounded variation.

Define

$$\hat{f}(\gamma) = \int_{-\infty}^{\infty} f(x) e^{i\gamma x} dx$$

We will suppose for simplicity that F is entire. Then choosing $T > 0$ and defining

$$f(x) = \begin{cases} 2T - |x| & \text{if } |x| \leq 2T, \\ 0 & \text{otherwise.} \end{cases}$$

Then f satisfies the conditions (i) and (ii) above. Moreover,

$$\hat{f}(\gamma) = \left(\frac{2 \sin \gamma T}{\gamma} \right)^2.$$

Without difficulty, we derive the explicit formula,

$$\sum_{\infty} \hat{f}(\gamma) = f(1) \log Q - \sum_{p^k \leq e^T} \frac{b_{p^k}(F) f(\log p^k) + \overline{b_{p^k}(F)} f(-\log p^k)}{p^{k/2}} + O(T)$$

where the latter O-term represents the contribution from the Γ -factors in the functional equation, and the left hand side is over zeroes of $F(s)$ of the form $s = \frac{1}{2} + i\gamma$, $0 \leq \text{Re } s \leq 1$.

We now follow the method of [M3] with a slight variation. Observe that if $s = \frac{1}{2}$ and we choose $T = \log x$,

$$\hat{f}(0) = 4(\log x)^2$$

The left hand side of the explicit formula will contribute

$$4n(F)(\log x)^2$$

if $n(F)$ is the order of the zero $F(s)$ at $s = \frac{1}{2}$.

4. The main theorem

Theorem. *Let $F, G \in \mathcal{S}$ be such that $a_p(F) = a_p(G)$ for all but finitely many primes p . Suppose that F and G satisfy the quadratic twist hypothesis. That is, there exists a fundamental discriminant D such that*

$$\operatorname{ord}_{s=1/2} F_D(s) = \operatorname{ord}_{s=1/2} G_D(s)$$

Then $F = G$.

Proof. For simplicity, we shall assume F and G are entire. The modification of the proof below to treat the general case is minor. In fact, as indicated below, the zeroes and poles of F in the region $\operatorname{Re}(s) \geq \frac{1}{2}$ coincide with those of G under the hypothesis $a_p(F) = a_p(G)$. Their contribution to the explicit formula as used below is therefore nil.

We first note that $F(s)/G(s)$ is analytic in $\operatorname{Re}(s) > 1/2$. Thus, any zero β of F not lying on $\operatorname{Re}(s) = 1/2$ but with $\operatorname{Re}(\beta) > 1/2$ coincides (with multiplicity) with a zero of G with the same property. Moreover, the contribution of zeroes β of with $\beta \neq 1/2$ and $\operatorname{Re}(\beta) = 1/2$ is $O(1)$. Taking into account the contribution from the possible zero at $s = 1/2$ gives us the equation

$$\begin{aligned} & 4(n(F) - n(F_D))(\log x)^2 - 4(n(G) - n(G_D))(\log x)^2 \\ &= \sum_{p^k \leq x} \frac{b_{p^k}(F) + b_{p^k}(\overline{F})}{p^{\frac{k}{2}}} \left(1 - \left(\frac{D}{p^k}\right)\right) (\log p^k) (\log \frac{x}{p^k}) \\ & - \sum_{p^k \leq x} \frac{b_{p^k}(G) + b_{p^k}(\overline{G})}{p^{k/2}} \left(1 - \left(\frac{D}{p^k}\right)\right) (\log p^k) (\log \frac{x}{p^k}) + O(\log x) \end{aligned}$$

for any fundamental discriminant D . Since $n(F_D) = n(G_D)$ and the contribution from the terms p^k with $k \geq 3$ is $O(\log x)$, we see that the terms in the sum with $k = 1$ and $k = 2$ disappear. This is because for $k = 1$, $b_p(F) = b_p(G)$ by hypothesis and for $k = 2$,

$$\left(\frac{D}{p^2}\right) = 1$$

Hence,

$$4(n(F) - n(G))(\log x)^2 = O(\log x)$$

Since x is arbitrary, we let $x \rightarrow \infty$ and deduce that $n(F) = n(G)$. The same argument can now be applied to $F(s + i\gamma_0)$ where $\frac{1}{2} + i\gamma_0$ is a zero of $F(s)$. The argument then leads to

$$\operatorname{ord}_{s=\frac{1}{2}+i\gamma_0} F(s) = \operatorname{ord}_{s=\frac{1}{2}+i\gamma_0} G(s)$$

Hence all the zeroes of $F(s)$ and $G(s)$ on $\operatorname{Re} s = \frac{1}{2}$ coincide with the multiplicity.

This completes the proof of the Theorem.

References

- [B] S. Bochner, On Riemann's functional equation with multiple gamma factors, *Annals of Math.* 67 (1958) p.29-41
- [CG] B. Conrey and A. Ghosh, On the Selberg class of Dirichlet series: small degrees, *Duke Math. Journal*, 72 (3) (1993) p. 673-693.
- [M1] M. Ram Murty, Selberg's conjectures and Artin L -functions, *Bulletin of the A.M.S.*, 31(1) (1994) p. 1-14.
- [M2] M. Ram Murty, Selberg's conjectures and Artin L -functions, II, in *Current Trends in Mathematics and Physics, A tribute to Harish-Chandra*, (edited by S.D. Adhikari) Narosa Publishing House, 1995.
- [M3] M. Ram Murty, Simple zeroes of L -functions in Number Theory (edited by R. Mollin) (1989) p. 427-439. Walter de Gruyter.
- [L] S. Lang, *Algebraic Number Theory*, Springer-Verlag, (Second edition), 1994.
- [MM] M. Ram Murty and V. Kumar Murty, Strong Multiplicity one for Selberg's class, *C.R. Acad. Sci. Paris*, t.319, Série I, (1994) p.315-320.
- [MM2] M. Ram Murty and V. Kumar Murty, Mean values of derivatives of modular L -series, *Annals of Math.*, 133(1991) p.447-475.
- [S] A. Selberg, Old and new conjectures and results about a class of Dirichlet series, in *Collected Papers, Volume 2*, Springer-Verlag, Berlin (1991), p. 47-63.

Department of Mathematics
 McGill University
 Montreal (Que)
 CANADA H3A 2K6